Arista Access Node Failure Handling Procedure (ALARM_8339)

Node: BAS-ACC-78 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_8339

Description

The Access Node failure domain in the Arista BAS-ACC-78 equipment, located at the Basavanagudi POP, is a result of Hardware/Environmental Faults such as ASIC memory parity errors, line card crashes, or power supply redundancy loss. In such scenarios, the Arista equipment generates a Warning alarm code (ALARM_8339) due to the loss of redundancy in the power supply or line card failures, leading to potential packet loss, throughput degradation, or even node isolation. The ASIC memory parity errors can occur due to hardware defects, aging, or thermal issues. The line card crashes may be attributed to software bugs, configuration inconsistencies, or incompatibility issues. Power supply redundancy loss can be caused by a faulty power supply unit, improper configuration, or environmental factors such as power grid fluctuations. To prevent such failures and ensure network resilience, it is crucial to implement regular hardware and software maintenance, monitor equipment temperatures, and perform thorough configuration verifications.

Pre-requisite Checks

- Verify the Basavanagudi POP site location and ensure remote hands are dispatched to investigate.

- Check the redundant link status and confirm that the backup links are functioning properly.

- Verify the power supply units and confirm that they are functioning within normal parameters.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware defects or thermal issues	High
Line card crashes due to software bugs or configuration inconsistencies	Medium
Power supply redundancy loss due to faulty power supply units or improper configuration	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal hardware cpu-asic detail, Expected Output: asic memory parity errors detected, Description: This command displays detailed information about the ASIC memory parity errors, including the error count and the affected ASICs. This helps in isolating the root cause of the failure.

Command: show system internal linecard detail, Expected Output: linecard crash detected, Description: This command displays detailed information about the line card crashes, including the crash time and the affected line card. This helps in identifying the root cause of the failure.
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```
Command: show system internal power-supply detail, Expected Output: power supply redundancy loss detected, Description: This command displays detailed information about the power supply redundancy loss, including the affected power supply units. This helps in identifying the root cause of the failure.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: no system internal hardware cpu-asic replace , Description: Replace the faulty ASIC with a new one to resolve the ASIC memory parity errors. This step requires remote hands to physically replace the ASIC.

Step 2: Step: Restart the line card, Command: restart system internal linecard , Description: Restart the line card to resolve the line card crashes. This step helps in clearing any software bugs or configuration inconsistencies.

Step 3: Step: Replace the faulty power supply unit, Command: no system internal power-supply replace , Description: Replace the faulty power supply unit with a new one to resolve the power supply redundancy loss. This step requires remote hands to physically replace the power supply unit.

Post-Check Verification

```
Command: show system internal hardware cpu-asic detail, Expected Output: no asic memory parity errors detected, Description: Verify that the ASIC memory parity errors have been resolved by running this command.

Command: show system internal linecard detail, Expected Output: no linecard crash detected, Description: Verify that the line card crashes have been resolved by running this command.

Command: show system internal power-supply detail, Expected Output: no power supply redundancy loss detected, Description: Verify that the power supply redundancy loss has been resolved by running this command.
```

Roll-back Procedures

- Step: Restore the original ASIC configuration, Command: system internal hardware cpu-asic restore , Description: Restore the original ASIC configuration in case the fix fails.

- Step: Restart the line card, Command: restart system internal linecard , Description: Restart the line card in case the fix fails.

- Step: Restore the original power supply configuration, Command: system internal power-supply restore , Description: Restore the original power supply configuration in case the fix fails.